AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 9 · STUDENT LESSON

Slopes and local change

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Sample nearby. Divide by the full gap. Check the move.

Estimate local slopes while separating direction clues from global or step-size guarantees.

Keep one habit: Complete quadratic table → symmetric estimates → pictured overshoot → labelled zero-slope counterexample → independent new curve.

MAIN LESSON · SECTION 1

Which small change might lower the loss here?

> A slope is a local direction clue,not a promise about every jump.

Learning goals

I can calculate an average rate of change over a stated interval.

I can estimate a local **slope** with nearby loss values.

I can explain the limits of a local direction clue.

Retrieve loss and parameters

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- For one example with **prediction** 4 and target 2, what is the squared error?

> **My starting explanation**
>
> _____
>
> _____
>
> _____
>
> _____

End goal

By the end, I can explain **slope**s and **local change** using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Calculate an average rate of change over a stated interval

Estimate a local **slope** with nearby loss values

Explain the limits of a local direction clue

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Vary one parameter and record its loss

> The horizontal coordinate is a parameter,not a new data example.

Declare the one-example model and squared loss

Let a one-example model predict w and let its target be 2. Its squared loss is L(w) = (w - 2) squared. Here w is the parameter we vary, not a new data example. L is a nonnegative function of that parameter.

Complete parameter/loss table

w	Prediction minus target	L(w)
0	-2	4
1	-1	1
2	0	0
3	1	1
4	2	4

Compute average slopes on named intervals

From w = 0 to w = 1, the loss changes by -3 while w changes by 1. The average **slope** is -3.
From w = 1 to w = 2 it is -1. The curve does not have one fixed **slope** everywhere.

Name given and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Divide the loss change by the full horizontal gap

> Keep the centre and both nearby evaluations explicit.

Complete near-w1calculation

Near w = 1, use two symmetric nearby values: 0.9 and 1.1. Their losses are 1.21 and 0.81.
Change in loss divided by change in w gives (0.81 - 1.21) / (1.1 - 0.9) = -2.

This symmetric estimate describes change near the centre w = 1.

Four complete symmetric estimates

Centre w / symmetric step h · L(w–h) · L(w+h) · Loss difference · Parameter difference 2h · Estimated local slope

> **Centre w / symmetric step h:** 0 /0.5
>
> **L(w–h):** 6.25

L(w+h): 2.25

Loss difference: -4

Parameter difference 2h: 1

Estimated local slope: -4

Centre w / symmetric step h: 1 /0.1

L(w−h): 1.21

L(w+h): 0.81

Loss difference: -0.4

Parameter difference 2h: 0.2

Estimated local slope: -2

Centre w / symmetric step h: 2 /0.1

L(w−h): 0.01

L(w+h): 0.01

Loss difference: 0

Parameter difference 2h: 0.2

Estimated local slope: 0

Centre w / symmetric step h: 3 /0.1

L(w−h): 0.81

L(w+h): 1.21

Loss difference: 0.4

Parameter difference 2h: 0.2

Estimated local slope: 2

Spacing,denominator and precision contract

The symmetric formula is[L(w+h)−L(w−h)]/(2h), centred at w. Its h is a measurement spacing for estimating slope, not a training learning rate. At w1, h.1, the full horizontal gap is.2, not.1. For this quadratic the symmetric estimate matches 2(w−2)in exact arithmetic; other functions generally have approximation error. Extremely small spacing can also lose numerical precision.

Record centre, spacing, both losses and the full denominator.

For this quadratic, the symmetric estimate equals the exact **slope** 2(w - 2) in exact arithmetic. Numerical estimates for other functions are generally approximate.

Smaller steps can reduce some approximation errors, but extremely small steps can suffer from computer rounding.

Runnable positive-spacing estimator

```
def loss(w):
    return (w - 2) ** 2

def local_slope(w, step):
    if step <= 0:
        raise ValueError("Step must be positive")
    return (loss(w + step) - loss(w - step)) / (2 * step)

for w in [1, 2, 3]:
    print(w, round(local_slope(w, 0.1), 6))
# Slopes near these centres: -2, 0, 2
```

Connect visual values to the calculation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare a local clue with the loss after a proposed move

> Estimate,propose and then check;do not skip the new evaluation.

Current symmetric-estimate task

- At centre w = 0, use step 0.5. Calculate L(-0.5), L(0.5) and the symmetric **slope**.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured jump to5

- Change: Move from 0 all the way to5.

▼ **Check after predicting**

Prediction: Loss becomes 9, exceeding the starting 4.

Reason: A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

See a small successful move and a large failed move

> Both changes increase the parameter;only the new losses settle what happened.

Complete picture facts and letter/fixture boundary

Use a local **slope** to propose a small move rather than claiming every move helps.

Toy loss $J(a)=(a-2)^2$. At a = 0, use symmetric step 0.5.

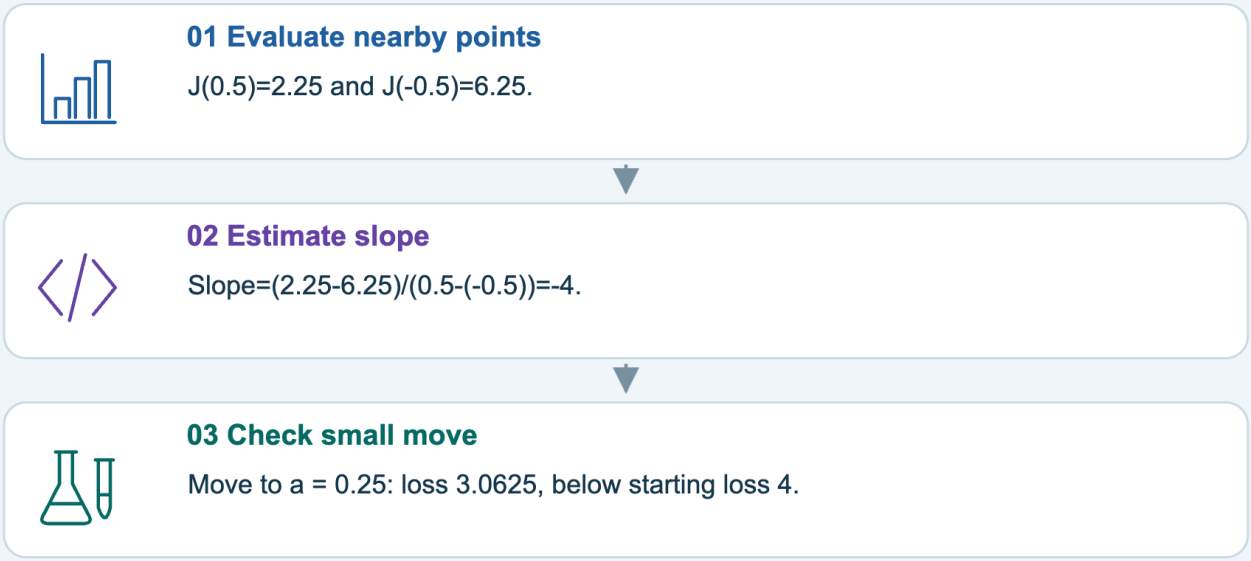
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture writes the same main quadratic with letters $J(a)=(a-2)^2$. At a0, h.5:nearby losses 6.25 and2.25 give slope−4. A small increase to.25 gives3.0625, below starting 4; the jump to5 gives9, above 4. The local negative slope supports testing a small increase, not any increase. The independent exercise uses $J(a)=(a+1)^2$ instead; do not transfer the pictured optimum or numbers to it.

Trace nearby values and small-move check

AI ACADEMY / AI-5/9 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Evaluate nearby points	J(0.5)=2.25 and J(-0.5)=6.25.
2. Estimate slope	Slope=(2.25-6.25)/(0.5-(-0.5))=-4.
3. Check small move	Move to a = 0.25: loss 3.0625, below starting loss 4.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **slope** describes how the loss changes near the current parameter. The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero.

It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

Predict the jump from0to5

- Change one thing: Move from 0 all the way to5. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-5/9 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Move to a = 0.25: loss 3.0625, below starting loss 4.

Changed condition

Move from 0 all the way to5.

Worked changed-case result

Loss becomes 9, exceeding the starting 4.
Why: A negative local slope suggests increasing a locally, not that every positive jump decreases loss.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the counterexample and local limits

- A finite difference approximates **local change**; numerical precision and nonsmooth points need care in more complex objectives.

- Read the labelled picture: Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the main near-w3estimate

Compare slope clues on the same loss curve

CURRENT W	LOCAL SLOPE	NEARBY DOWNHILL DIRECTION
1	-2	Increase w a little
2	0	No first order direction
3	2	Decrease w a little

These are local clues for L(w) = (w - 2) squared. At w = 2 the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

At w = 3, use 2.9 and 3.1. Calculate both losses and the symmetric slope. Which small change should lower loss here?

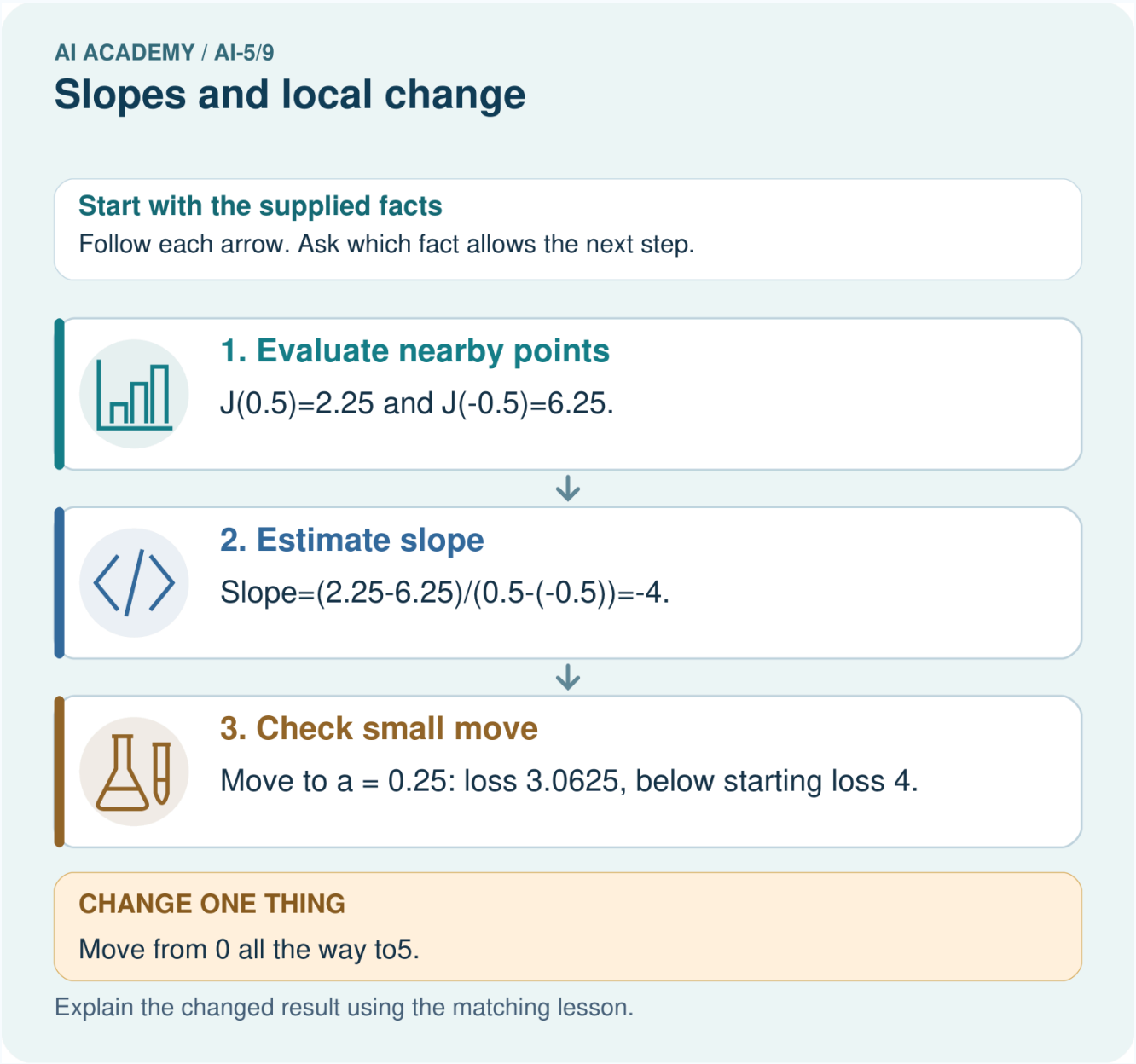
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Slopes and local change

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Repair “any step helps”and “zero slope means minimum”

> Use complete numerical counterexamples and labelled curves.

Small-step versus jump and zero-slope counterexamples

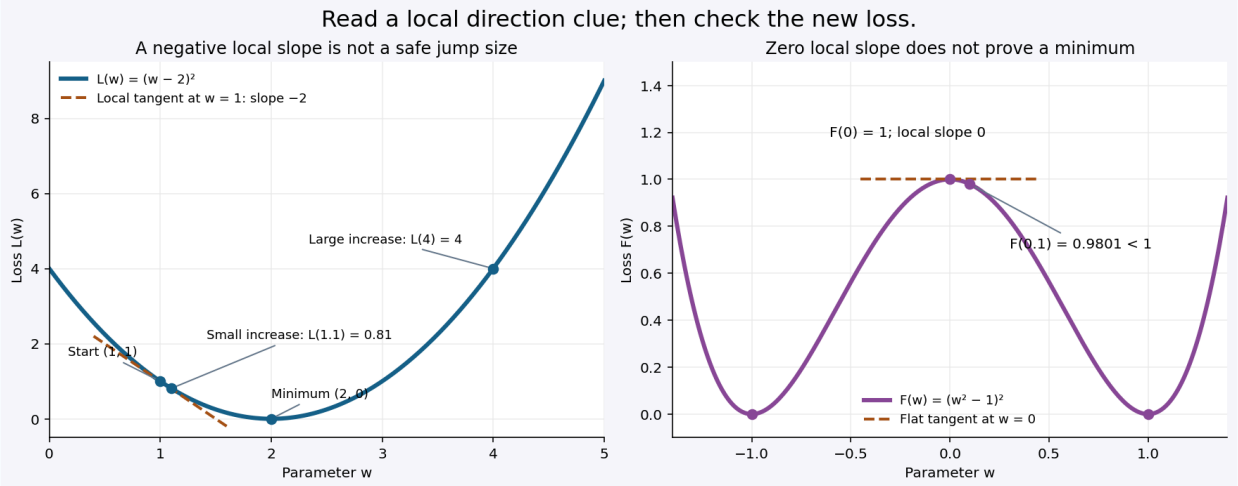
A negative **slope** at w = 1 suggests that a sufficiently small increase reduces this smooth loss locally. Moving to 1.1 lowers loss from 1 to 0.81. A jump all the way to 4 raises loss to 4.

The initial **slope** cannot justify a step of any size.

A zero **slope** is also limited **evidence**. The nonnegative function F(w) = (w squared - 1) squared has zero local **slope** at w = 0, yet F(0) = 1 and F(0.1) = 0.9801. Small moves on either side lower that function.

A flat first-order clue can occur at a point that is not a minimum.

Two labelled loss curves



Try both decreasing moves fromw3

- Starting at w = 3, compare loss after moving to 2.9 and after moving to -1. Both moves decrease w.

> **State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise differences,denominators and direction with less help

> State where the estimate applies before suggesting a change.

Current task and response

- For L(w) = (w - 2) squared, calculate the average **slope** from w = 0 to w = 2. Compare it with the local **slope** at w = 2.

> **At w = 3, use 2.9 and 3.1. Calculate both losses and the symmetric slope. Which small change should lower loss here?**

> **Explain your prediction to a partner before comparing results.**

Supported W2record

Use workbook W2 for the connected example “**Slope**s and **local change**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Slope**s and **local change**. Proposed explanation: “Divide by the step only”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured overshoot reasoning

- Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the independent new curve

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Check a symmetric estimate

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate sign,spacing,overshoot and a flat counterexample

All function definitions,centres and step sizes are supplied.

Current task and runnable estimator

At centre w = 0, use step 0.5. Calculate L(-0.5), L(0.5) and the symmetric **slope**.

```
def loss(w):
    return (w - 2) ** 2

def local_slope(w, step):
    if step <= 0:
        raise ValueError("Step must be positive")
    return (loss(w + step) - loss(w - step)) / (2 * step)
```

```
for w in [1, 2, 3]:
    print(w, round(local_slope(w, 0.1), 6))
# Slopes near these centres: -2, 0, 2
```

Complete normal,boundary,invalid and changed-function probes

Complete local-slope probe	Expected calculation / action
Normal centre 1, h.1	Losses1.21/.81, fullgap.2, slope−2
Positive slope centre 3, h.1	Losses.81/1.21, slope 2; test small decrease
Zero slope centre 2, h.1	Both nearby losses.01, slope 0; this specific quadratic also has minimum 0 there
Step0or−.1	ValueError:Step must be positive
Missing centreNone	Outside numeric input contract; request/validate before calculation
Small decrease 3→2.9 versuslarge3→−1	Loss1→.81 helps; loss 1→9 worsens
Different functionF(w)=(w²−1)²at0	Symmetric slope 0by symmetry, but F(.1)=.9801< F(0)=1
Average interval 0→1 versuslocalcentre1	Average slope−3, local quadratic slope−2; do not equate them

Predict,calculate,run or trace,and check

1. Predict each complete estimate and new loss before using the supplied loss/local_slope functions. Keep exact paper calculations beside rounded computer results. Use separate calls for invalid spacing cases. The code checks positive spacing but assumes numeric, finite centre/spacing; it is not a general validator. Check the counterexample Fseparately because local_slope is tied to the declared Lfunction.

   These arithmetic trials do not train a network or establish a globally safe step. Retain centre, both sampled points, denominator, new parameter and its checked loss.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: a new curve with a different optimum

> Use its own function,centre and spacing,then check the proposed moves.

Complete independent function and tasks

Let J(a) = (a + 1) squared. At a = 1 use step 0.25 to find the symmetric **slope**. Compare J(1) with J(0.9), then explain why moving to -4 does not follow safely from the **slope** clue.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Estimate a new curve independently

Attempt this without hints

Let J(a) = (a + 1) squared. At a = 1 use step 0.25 to find the symmetric **slope**. Compare J(1) with J(0.9), then explain why moving to -4 does not follow safely from the **slope** clue.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the function and local calculation with the proposed move

> A reader should see the evidence before and after the change.

Project artifact and evidence table

Add a labelled **slope**s and **local change evidence** sheet to Image-pattern Network Investigation. Show how you can calculate an average rate of change over a stated interval.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one local clue and its checked limit

> Explain why the denominator and jump size matter.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Calculate an average rate of change over a stated interval

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Slope	Change in output divided by change in input over a stated interval
Local change	Change considered near a particular input or parameter value
Step size	The amount by which an input or parameter is changed
Stationary point	A point where a differentiable function has zero local slope

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For L(w) = (w - 2) squared, calculate the average **slope** from w = 0 to w = 2. Compare it with the local **slope** at w = 2.
- Starting at w = 3, compare loss after moving to 2.9 and after moving to -1. Both moves decrease w.
- For F(w) = (w squared - 1) squared, compare F(0) and F(0.1). Does zero local **slope** at zero prove a minimum?

Explain → apply → challenge

Explain which evidence supports your conclusion and which claim remains unproved.

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent $J(a)=(a+1)^2$curve for the first individual attempt.

Current explanation and response

- Always record the function, centre, nearby values and denominator used in a **slope** estimate. An average **slope** across a wide interval is not automatically the local **slope** at either endpoint.

  Do not compare **slope** magnitudes without considering parameter and loss units.

- Next week changes one parameter at a time in a two-parameter model and combines those local **slope**s into a gradient.

- For F(w) = (w squared - 1) squared, compare F(0) and F(0.1). Does zero local **slope** at zero prove a minimum?

A learner sees a negative estimated slope and says any larger parameter value will improve the loss. Give a counterexample using this lesson.

My limited claim and a useful next check

▼ Extra practice · Revisit the complete trace

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Compare slope clues on the same loss curve

CURRENT W	LOCAL SLOPE	NEARBY DOWNHILL DIRECTION
1	-2	Increase w a little
2	0	No first order direction
3	2	Decrease w a little

These are local clues for L(w) = (w - 2) squared. At w = 2 the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Use a local **slope** to propose a small move rather than claiming every move helps.

By the end, I can	How I will show it
Calculate an average rate of change over a stated interval	Explain the deciding step, show a trace or test, and state one limit.
Estimate a local slope with nearby loss values	Explain the deciding step, show a trace or test, and state one limit.
Explain the limits of a local direction clue	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Use a local **slope** to propose a small move rather than claiming every move helps.
- Which labelled step matches this detail? Move to a = 0.25: loss 3.0625, below starting loss 4.
- 1. Check small move
- 2. Estimate **slope**
- 3. Evaluate nearby points
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check small move Move to a = 0.25: loss 3.0625, below starting loss 4. A **slope** describes how the loss changes near the current parameter.

The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero.

It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

Changed-case question and optional hint

- Changed-case practice: Move from 0 all the way to5. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Loss becomes 9, exceeding the starting 4.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Loss becomes 9, exceeding the starting 4. A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

Repair and retain the record

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is B positive under the original or changed model?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

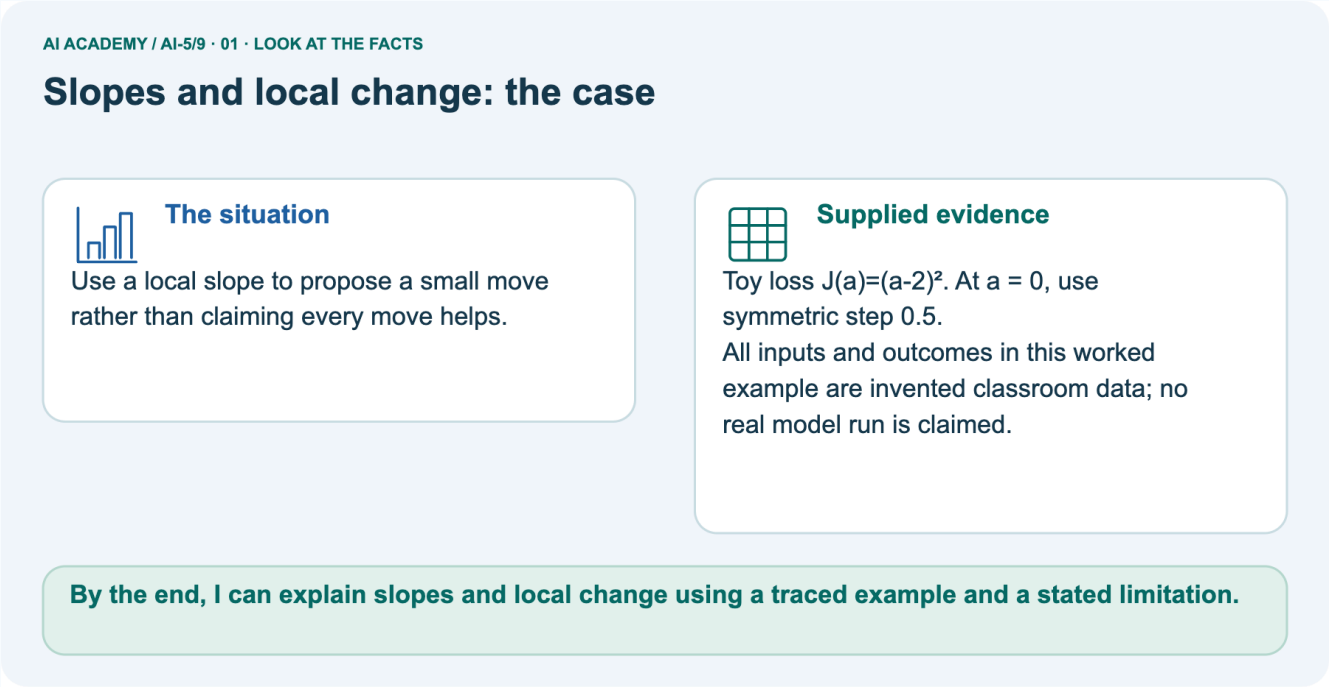
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Slopes and local change: the case



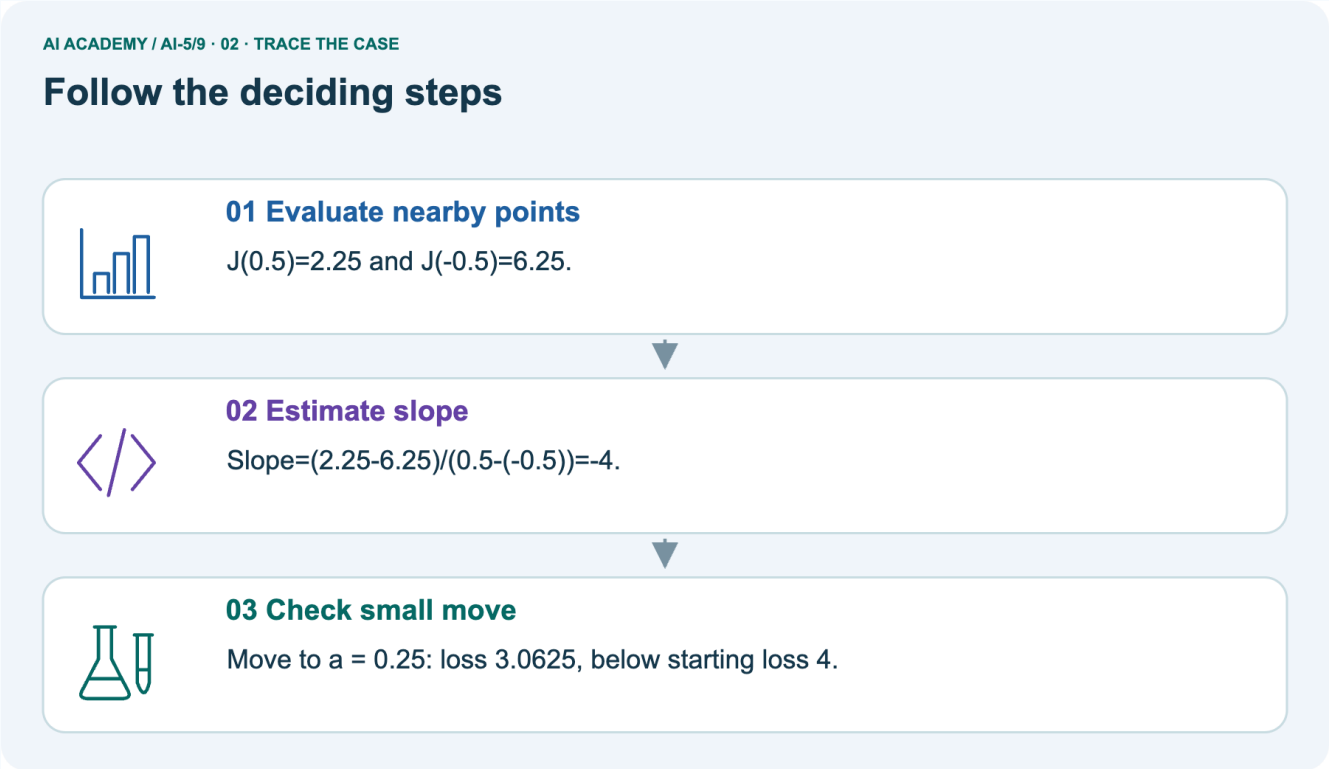
Follow the labelled picture with the matching explanation. Open the vector diagram

Use a local **slope** to propose a small move rather than claiming every move helps. Supplied facts: Toy loss J(a)=(a-2)². At a = 0, use symmetric step 0.5.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Evaluate nearby points: J(0.5)=2.25 and J(-0.5)=6.25. → Estimate **slope**: **Slope**=(2.25-6.25)/(0.5-(-0.5))=-4. → Check small move: Move to a = 0.25: loss 3.0625, below starting loss 4.

Point to each numbered stage and explain which recorded **input** it uses.

03. Compare slope clues on the same loss curve

AI ACADEMY / AI-5/9 · 03 · INSPECT THE LESSON MODEL

Compare slope clues on the same loss curve

Current w	Local slope	Nearby downhill direction
1	-2	Increase w a little
2	0	No first order direction
3	2	Decrease w a little

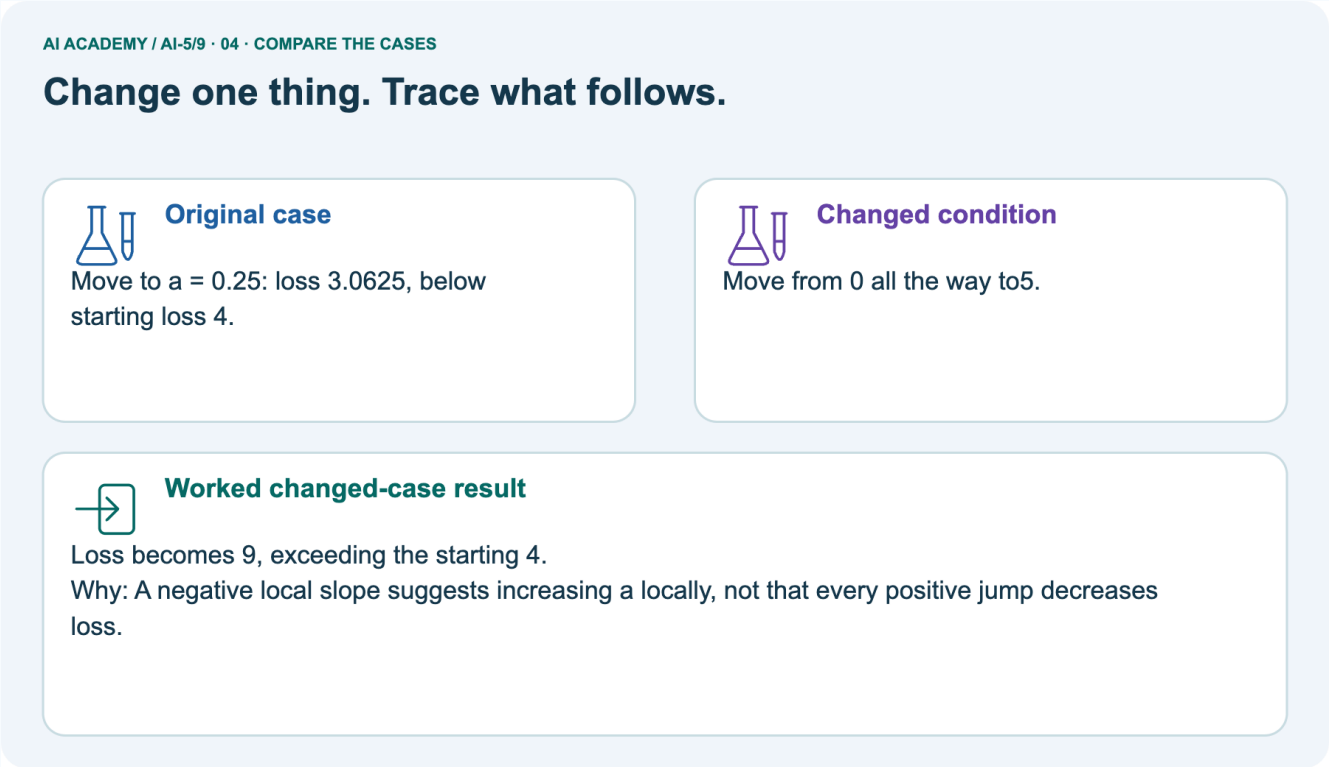
These are local clues for L(w) = (w - 2) squared. At w = 2 the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

Follow the labelled picture with the matching explanation. Open the vector diagram

Compare **slope** clues on the same loss curve. 1 | -2 | Increase w a little. 2 | 0 | No first order direction. 3 | 2 | Decrease w a little

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



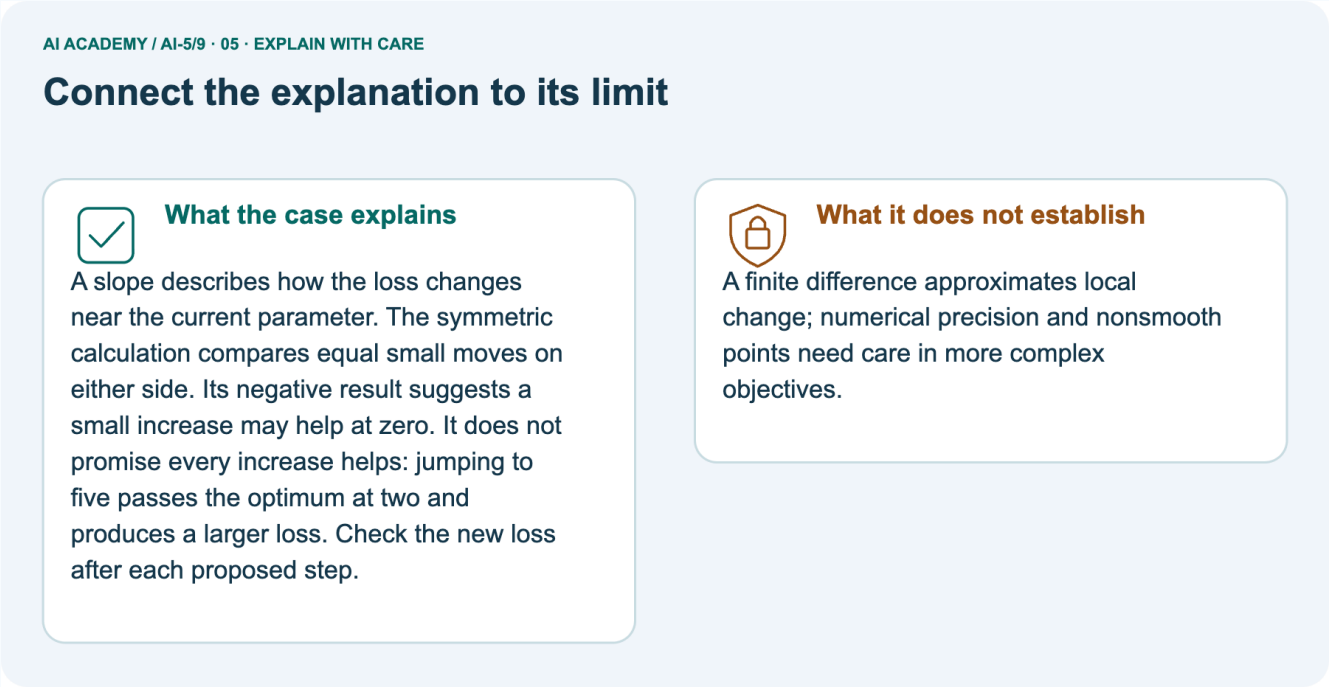
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Move to a = 0.25: loss 3.0625, below starting loss 4. Changed condition: Move from 0 all the way to5. Expected result: Loss becomes 9, exceeding the starting 4.

Reason: A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

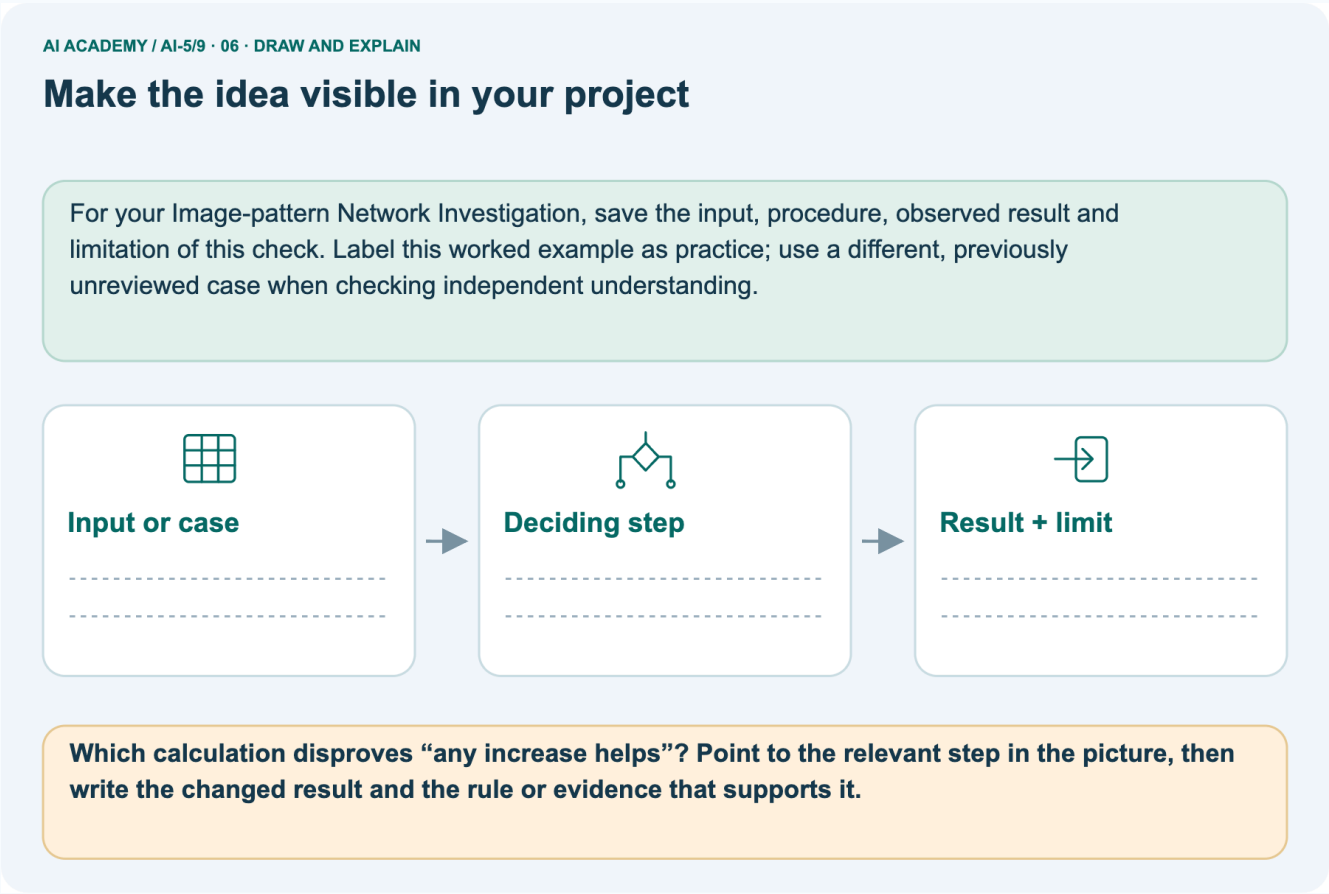
A **slope** describes how the loss changes near the current parameter. The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero.

It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

Limit: A finite difference approximates **local change**; numerical precision and nonsmooth points need care in more complex objectives.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Slopes and local change: the case

02. Follow the deciding steps

03. Compare slope clues on the same loss curve

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.